

On the Expansion of $(1 - \zeta_p)^n$

Student-Faculty Research

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ISMAA 2026

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- Old information is used
- Only need ζ_3^0 and ζ_3^1 to represent the binomial expansion.

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- How is the recurrence relation for the general prime number $(1 - \zeta_p)^n$?
- What is the runtime complexity using the recursive method?
 - When $p=3$, with a naive recursive approach, this is $\mathcal{O}(2^n)$ **arithmetic complexity**.

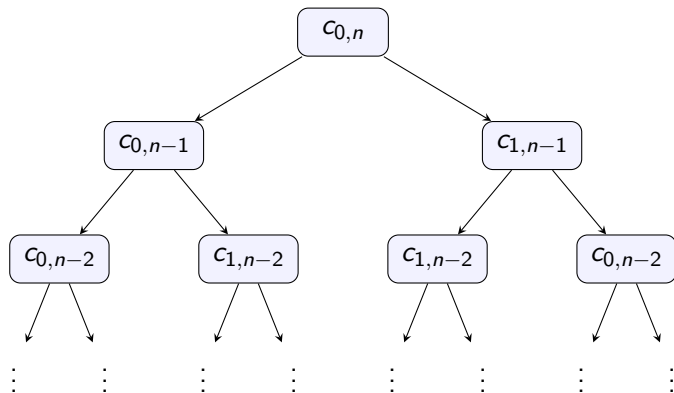


Figure: The naive binary recursive call tree for evaluating a single ROU coefficient at level n for $p = 3$. The execution tree doubles in width at every step, resulting in the severe $\mathcal{O}(2^n)$ arithmetic complexity and massive number of overlapping subproblems.

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- What is the runtime complexity using the recursive method?
 - When $p=3$, with a naive recursive approach, this is $\mathcal{O}(2^n)$ arithmetic complexity.
 - In general, this is $\mathcal{O}((p-1)^n)$
- Solve the original p -adic valuation problem

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Outline

- 1 Introduction
- 2 The Combinatorial Approach
- 3 The Trigonometric Approach
- 4 Computational Analysis
- 5 Summary
- 6 Future Work

Primitive Roots of Unity

(Root of Unity) Let p be a prime number. Then, the solutions to the equation $x^p = 1$, denoted as ζ_p , are called the p^{th} roots of unity (ROU).

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The p^{th} roots of unity can be calculated in the complex plane $\mathbb{C}$ as:

$$\zeta_p^k = \exp\left(\frac{2\pi k}{p}i\right),$$

where $i = \sqrt{-1}$, $k = 1, 2, \dots, p$.

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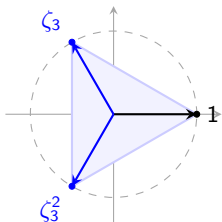
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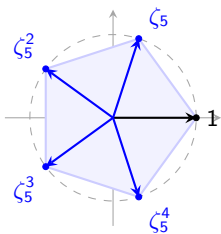
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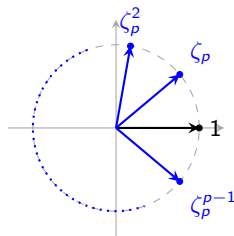
Meaning: We know there are exactly p roots for this equation $x^p = 1$, and we know what they are!



$p = 3$



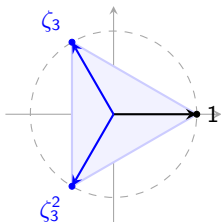
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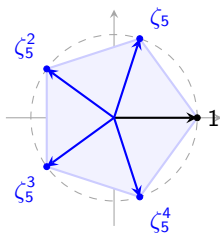
General p

Notice that, since $1 + \zeta_p + \zeta_p^2 + \cdots + \zeta_p^{p-1} = 0$, we can always express:

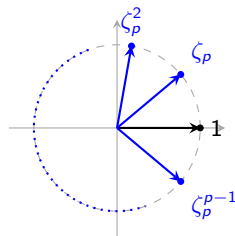
$$(1 - \zeta_p)^n = c_{0,n}\zeta_p^0 + c_{1,n}\zeta_p^1 + \cdots + c_{p-2,n}\zeta_p^{p-2} = \sum_{j=0}^{p-2} c_{j,n}\zeta_p^j$$



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Big question: What is $c_{j,n}$? Can we find the coefficient faster than $\mathcal{O}(2^n)$?

Binomial Theorem

(Binomial Theorem) For $a, b \in \mathbb{C}$, $n \in \mathbb{Z}_{\geq 0}$:

$$(a + b)^n = \sum_{k=0}^n \binom{n}{k} a^k b^{n-k}. \quad (1)$$

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(where are $\zeta_3^3, \zeta_3^4, \zeta_3^5$?)

Binomial Theorem (Cont.)

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The Combinatorial Expansion ($\mathcal{O}(n)$ Arithmetic Complexity)

By applying the Binomial Theorem and the property that $\sum_{k=0}^{p-1} \zeta_p^k = 0$, we conjecture that

Theorem: Combinatorial Expansion

For $n \geq 0$, the coefficients $c_{j,n}$ can be calculated as:

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Proof technique: Induction!

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- Instead of recalculating raw factorials from scratch, we use the incremental ratio:

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- This means we can just use the "old information" and step it forward using exactly one multiplication and one division.

Linear Arithmetic Complexity

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- **Step $k=1$** : $\binom{n}{p-1} \rightarrow \binom{n}{j+p} \rightarrow \binom{n}{2p-1}$, and so on...
- The whole process, in total, takes approximately n steps!

Quick Discussion (Cont.)

- The arithmetic complexity of this algorithm is $\mathcal{O}(n)$, fixing p . This might seem very counterintuitive, and you are right!
- This is $\mathcal{O}(n)$ arithmetic complexity if we compute the combination in a smart way.
- However:
 - We did not totally exploit the symmetric nature of the roots of unity.
 - Can we find a faster algorithm to compute the ROU coefficients?

Root of Unity Filter

Motivation: Find a connection between combination and root of unity.

Notice that:

$$\sum_{k=0}^{\lfloor (n-j)/p \rfloor} \binom{n}{j+pk} x^{j+pk} = \frac{1}{p} \sum_{k=1}^p (\zeta_p^k)^{-j} (1 + x\zeta_p^k)^n$$

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The sum over k acts as a filter such that:

$$\sum_{k=1}^p (\zeta_p^k)^m = \begin{cases} p & \text{if } p \mid m \\ 0 & \text{otherwise} \end{cases}$$

Root of Unity Filter: An Example

Let's look at $p = 3$. To simplify, let m be the exponent.

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Case 1: m is a multiple of p (e.g., $m = 3$)

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Case 1: m is a multiple of p (e.g., $m = 3$)

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The Trigonometric Derivation

The sketch is as follows:

$$c_{j,n} = \sum_{k=0}^{\lfloor n/p \rfloor} \left[(-1)^{pk+j} \binom{n}{j+kp} - (-1)^{pk+p-1} \binom{n}{p-1+kp} \right]$$

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The Trigonometric Expansion ($\mathcal{O}(\log n)$)

Finally, with some algebra and Euler's identity, we derived:

Theorem: Trigonometric Expression

For $0 \leq j \leq p-2$:

$$c_{j,n} = \frac{1}{p} 2^{n+2} \left[\sum_{k=1}^{(p-1)/2} \sin \left(\frac{k\pi}{p} (p-1-j) \right) \sin^n \left(-\frac{k\pi}{p} \right) \cos(\Phi) \right]$$

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where $\Phi = \frac{-2\pi k(p-1+j)+p\pi}{2p} + n \frac{\pi(2k+p)}{2p}$. This is $\mathcal{O}(\log n)$ arithmetic complexity.

Computational Analysis

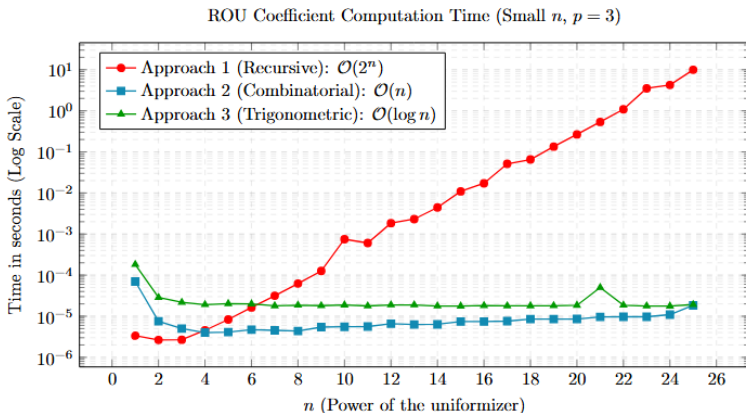
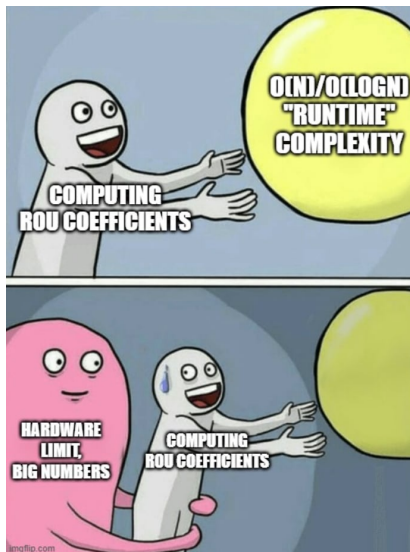


Figure: Performance comparison of the three enumeration formulas for $(1 - \zeta_p)^n$ at small values of n .

A Fun and Weird Hardware Limit



Computational Analysis (Cont.)

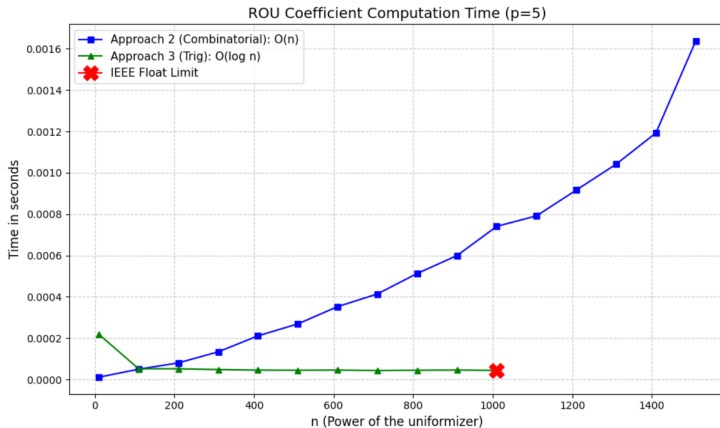


Figure: While the combinatorial formula maintains precision through linear time complexity, the $O(\log n)$ trigonometric formula triggers an IEEE floating-point overflow due to the exponential scalar before n reaches 1100.

Summary

- Found 2 closed-form formula to compute the ROU coefficients, which will have application to evaluate the p-adic problem
- Increased runtime to generate data for the coefficients.

- Study further runtime complexity
- Look at $(1 - \zeta_p)^n$ geometrically and through the lens of an eigenvalue-eigenvector problem.
 - $1 - \zeta_p = 2 \sin \left(\frac{\pi}{p} \right) \exp \left(i \left(\frac{\pi}{p} - \frac{\pi}{2} \right) \right)$
- **Conjecture:** The coefficients of the expansion $(1 - \zeta_p)^n$ repeat every $p(p-1)$ terms.
 - If proven, finding the maximum p -adic valuation k (where $c_{j,n} = p^k r$) shifts from an infinite parameter search to checking a finite bounded set.
 - These new $\mathcal{O}(n)$ and $\mathcal{O}(\log n)$ algorithms give us the computational power to gather the empirical data needed to prove this.

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Thank You!

Questions and Answer

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brookerandazzo@augustana.edu



Feedback or Ideas?

Proof of the Incremental Ratio

Theorem: For integers $n \geq 0$ and $0 \leq m < n$, $\binom{n}{m+1} = \binom{n}{m} \frac{n-m}{m+1}$

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Induction Proof Sketch

- Base Case: the case $n = 0$ is checked.
- Induction Step:
 - Induction Hypothesis: We assume that for $m \geq 0$:

$$(1 - \zeta_p)^m = \sum_{j=0}^{p-2} c_{j,m} \zeta_p^j.$$

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- We must show that this hypothesis applied for $(1 - \zeta_p)^{m+1}$. We will use our familiar trick.

Induction Proof Sketch

$$(1 - \zeta_p)^{m+1} = \underbrace{(1 - \zeta_p)^m}_{\text{apply IH}} \cdot (1 - \zeta_p)$$

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We want to show that

$$C_{j,m+1} = c_{j,m} - c_{j-1,m} + c_{p-2,m}.$$